IsDB-BISEW

TSP: U.S Software Limited

GARMENTS MANAGEMENT SYSTEM

(GMS ERP)

COMPREHENSIVE PROJECT PROPOSAL & DATABASE DESIGN

Document Information	Details
Prepared By	**JAKIR HOSEN LITON**
Trainee ID	1296764
Program	Is-DB-BISEW
TSP	U.S Software Limited
Course / Trade	Web Application Development
Batch / Round	WDPF/USSL-M/72/01
Instructor	Mohammad Towhidul Islam
Consultant/ Reviewer	Md. Moshaidul Islam
Project Type	**Garments Management System (ERP)**
Proposal Submission Date	03-11-2026
Planned Project Duration	30 days

Prepared for academic review and trainee project implementation.

1. Executive Summary

This proposal presents a garment-factory ERP blueprint covering merchandising, procurement, inventory, production, quality, human resources, finance and shipment. The database baseline is a 98-table MySQL/MariaDB schema. The proposal separates the complete system vision (18 modules) from the practical 30-day trainee scope: a connected, end-to-end workflow built with Core PHP and MySQL/MariaDB. It does not claim to deliver every advanced factory automation feature within one month.

1.1 Project Vision

- One connected database for buyer orders, styles, materials, purchasing, stock, production, QC, employees, accounts and delivery.
- Traceability from buyer PO and style through order, BOM, material issue, production output, inspection and shipment.
- Role-based access, validated CRUD, transaction-safe updates and useful operational reports.

1.2 Delivery Strategy

Layer	What it includes	Commitment
Complete ERP blueprint	18 modules and the full 98-table database	Documented in this proposal
30-day trainee MVP	All 18 modules at essential level: CRUD, validation, key calculations and reports	Implemented and demonstrated
Future enhancement	Statutory payroll, leave/shift rules, depreciation, advanced planning, integrations	Documented only

2. Project Objectives

- Build a database-driven Garments ERP using Core PHP, MySQL/MariaDB, HTML, CSS, Bootstrap and JavaScript.
- Implement secure login/logout, password hashing, sessions and role/department-based access.
- Manage buyers, styles, costing, samples, buyer POs, orders, products, BOM, suppliers, purchases and stock movements.
- Track production orders, stage output, QC inspection, employee attendance and basic payroll/asset records.
- Record invoices, receipts, income, expenses and shipment status; provide a dashboard and filtered reports.
- Deliver database documentation, ERD, test cases and a final demonstration.

3. Proposed ERP Modules (18 Modules)

The ERP consists of 18 modules covering the operational, administrative, financial and management functions of an export-oriented garments factory. Related functions are grouped so each module maps to a small set of database tables. The last column shows the planned implementation day(s) within the 30-day schedule.

#	Module	Main Functions	Priority
1	Dashboard	Summary cards, key statistics, recent orders, operational snapshots.	MVP
2	User & Role Management	Login/logout, user accounts, roles, department-wise access, protected pages.	MVP
3	Company & Department Master	Company profile, departments, positions, location reference data.	MVP – basic
4	Buyer Management	Buyer profile, contact person, email, phone, address, status.	MVP
5	Customer Management	Customer information, kept distinct from export buyers where needed.	MVP – basic
6	Merchandising Management	Style records, costing, sample status, buyer POs, size/color breakdown.	MVP
7	Product & Order Management	Order header and items, quantity, unit price, order and delivery dates.	MVP
8	Product Catalogue & BOM	Products, categories, units (UOM), BOM header and raw-material components.	MVP – basic
9	Purchase Management	Suppliers, purchase orders, purchase items, totals, supplier payments.	MVP
10	Inventory Management	Warehouses, stock-in/out, adjustments, balance, transaction history.	MVP
11	Production Management	Production orders, target/output quantity, stage-wise progress.	MVP
12	Quality Control (QC)	Inspection records, defect quantity, pass/fail status, remarks.	MVP
13	HR Management	Employee profile, department, designation, daily attendance.	MVP – attendance
14	Payroll Management	Payroll items, per-employee payroll invoice and details (basic).	MVP – basic

#	Module	Main Functions	Priority
15	Accounts Management	Invoices, money receipts, income/expense entries, outstanding balance.	MVP – basic
16	Fixed Asset Register	Asset name, purchase value and acquisition date (basic register).	MVP – basic
17	Shipment Management	Delivery, carrier, tracking number, dispatch date, delivery status.	MVP – basic
18	Reports	Order, stock, production, QC, HR, shipment and accounts reports with filters.	MVP – key reports

3.1 Thirty-Day MVP Boundary

Every module is delivered at an essential level so that one connected workflow can be demonstrated. Payroll and Fixed Asset are limited to basic records. Statutory payroll, leave entitlement, shift automation, asset depreciation, barcode/RFID, machine integration, multi-factory, multi-currency and external APIs are outside the initial implementation.

4. Main Business Workflows

Workflow	High-Level Sequence
Merchandising to PO	Buyer → Style → Costing / Sample → Buyer Purchase Order → Size/Color Breakdown
Order to Shipment	Order → Production Order → Production Stages → QC → Delivery → Shipment Tracking
Procurement to Stock	Supplier → Purchase → Purchase Details → Warehouse → Inventory Transaction
BOM to Production	Product → BOM → Raw Materials → Production → Stage Output → QC
HR to Attendance	Department / Position → Person → Employee Profile → Daily Attendance
Invoice to Receipt	Customer / Order → Invoice → Money Receipt → Outstanding Balance

5. System Architecture & Technology Stack

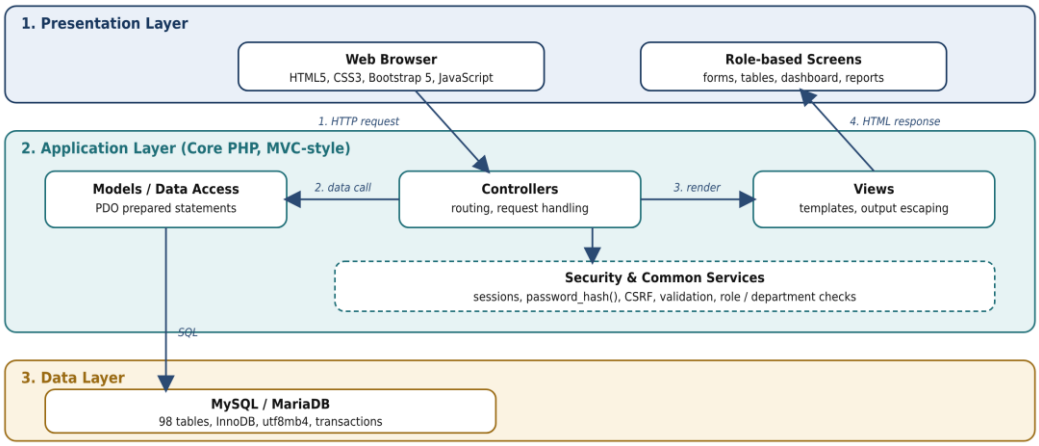


Figure 1: Three-layer, MVC-style architecture. Controllers handle requests, models use MySQLi prepared statements to interact with the MySQL/MariaDB database, and views return escaped HTML.

Layer	Proposed Technology / Practice
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript (developed on Windows, XAMPP, VS Code, Git/GitHub if permitted)
Backend	Core PHP; reusable functions/classes; server-side validation
Database	MySQL / MariaDB; InnoDB; utf8mb4; MySQLi prepared statements; transactions for multi-step updates
Security	PHP sessions, password_hash()/password_verify(), role checks, output escaping, CSRF protection, least-privilege DB user

6. Database Design & ERD

The database is the supplied 98-table schema (operational, master-data and technical tables). The ERD below is presented part by part, following the 18 modules, and shows only the key tables and columns of each module so that every diagram stays readable. Technical framework tables (cache, jobs, sessions, migrations, tokens) are not drawn.

98	98	69	17	core_	InnoDB
Tables	Primary keys	Declared foreign keys	Unique indexes	Table prefix	Target engine

How to read the diagrams: PK = primary key, FK = foreign key, UQ = unique. A solid line is a foreign key declared in the SQL; a dashed line is a logical relationship that should be enforced after data validation (39 recommended constraints). Grey boxes are tables of another module shown only for reference. The crow's-foot end is the "many" side.

6.1 Module Relationship Overview (18 Modules)

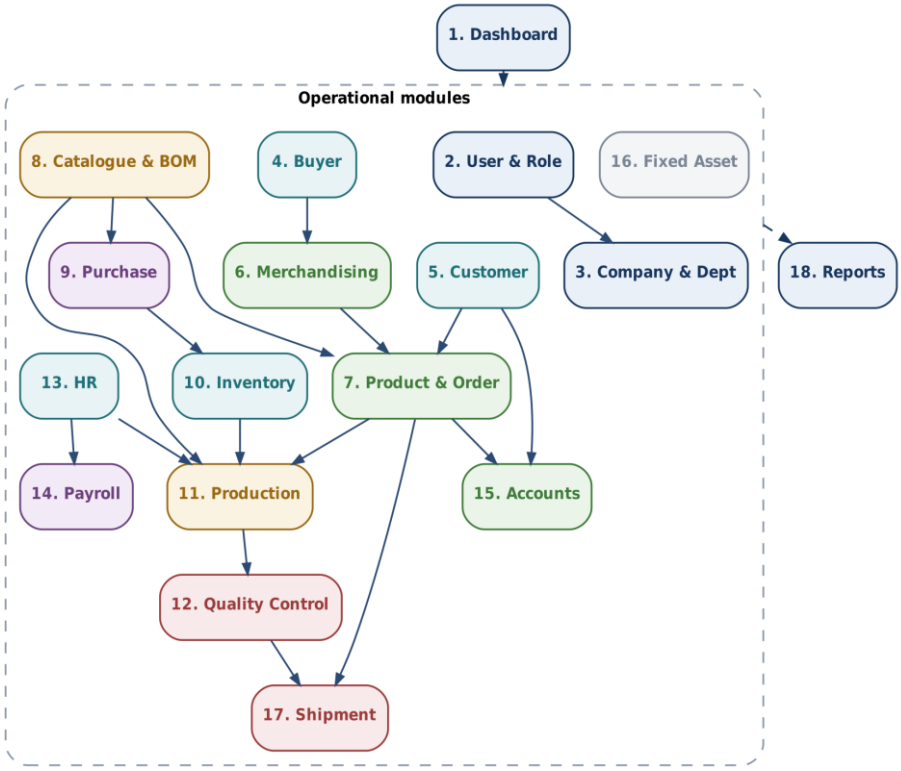


Figure 2: How the 18 modules connect. Dashboard and Reports read data from all operational modules.

6.2 Part 1 – Modules 2 & 3: Access Control and Company Master

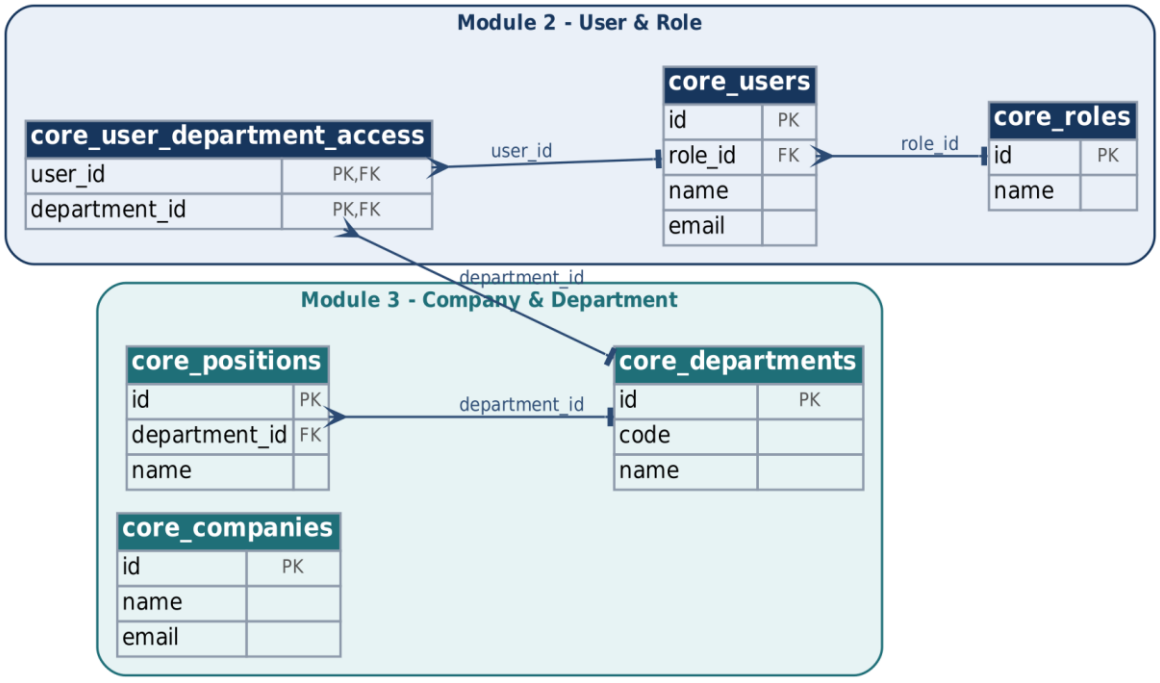


Figure 3: Users belong to a role; department access links users and departments (many-to-many); positions belong to departments.

6.3 Part 2 – Modules 4 & 6: Buyer and Merchandising

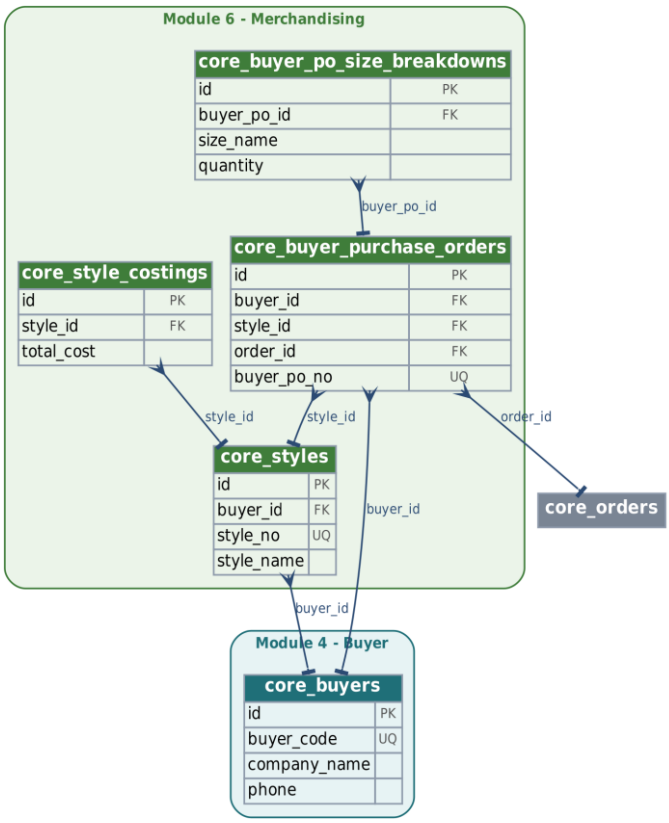


Figure 4: A buyer owns many styles; a style has costings; a buyer PO links buyer, style and order and is split into size/color quantities. Style samples follow the same style link as costings.

6.4 Part 3 – Modules 7 & 8: Product & Order, Catalogue & BOM

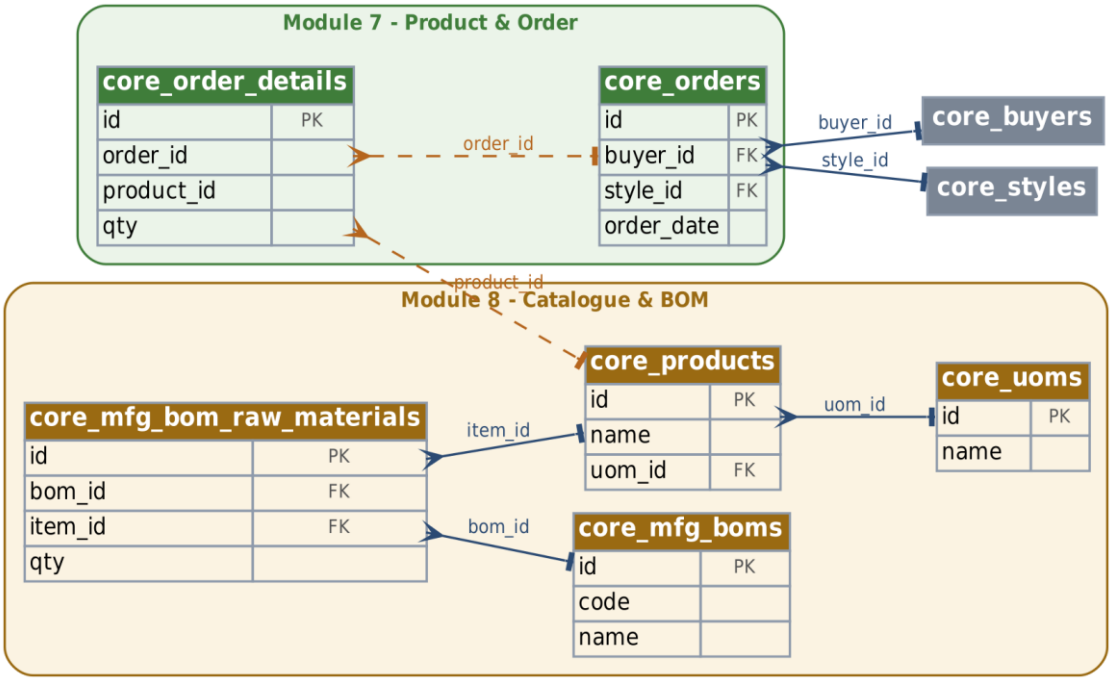


Figure 5: An order has many order lines; products use a unit of measure; a BOM lists its raw-material products.

6.5 Part 4 – Modules 9 & 10: Purchase and Inventory

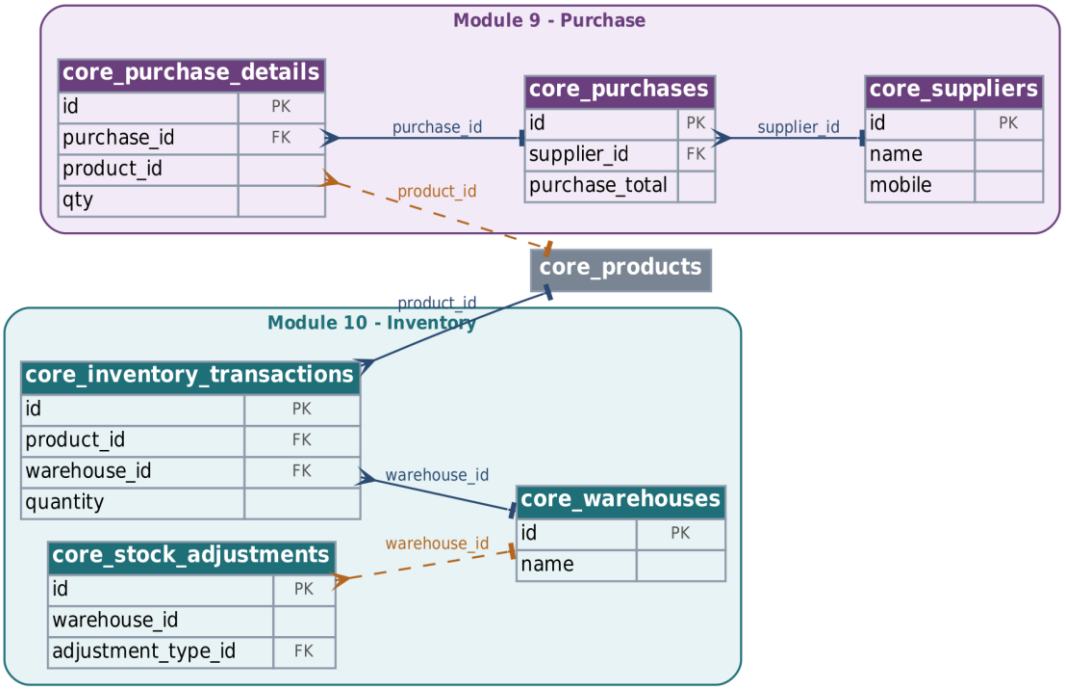


Figure 6: A supplier has many purchases and each purchase has items; every stock movement records a product and a warehouse.

6.6 Part 5 – Modules 11 & 12: Production and Quality Control

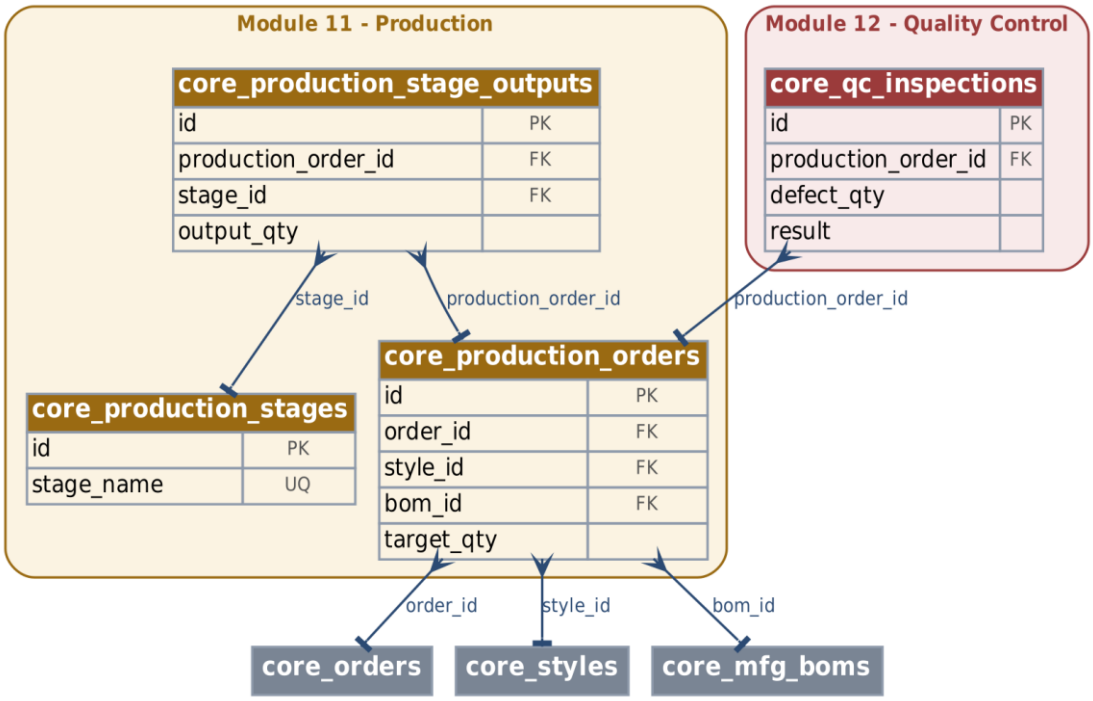


Figure 7: A production order is linked to order, style and BOM; stage output and QC inspection both reference the production order.

6.7 Part 6 – Modules 13 & 14: HR and Payroll

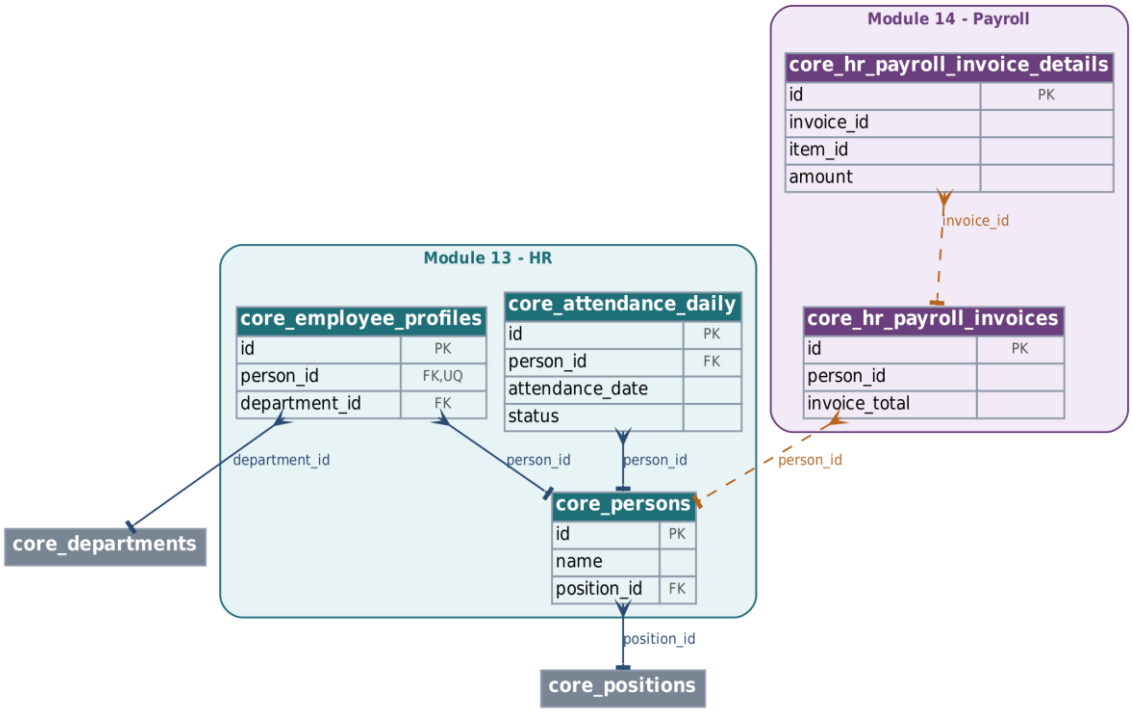


Figure 8: A person has one employee profile and many daily attendance rows; payroll invoices and their details belong to a person.

6.8 Part 7 – Modules 5, 15 & 16: Customer, Accounts and Fixed Assets

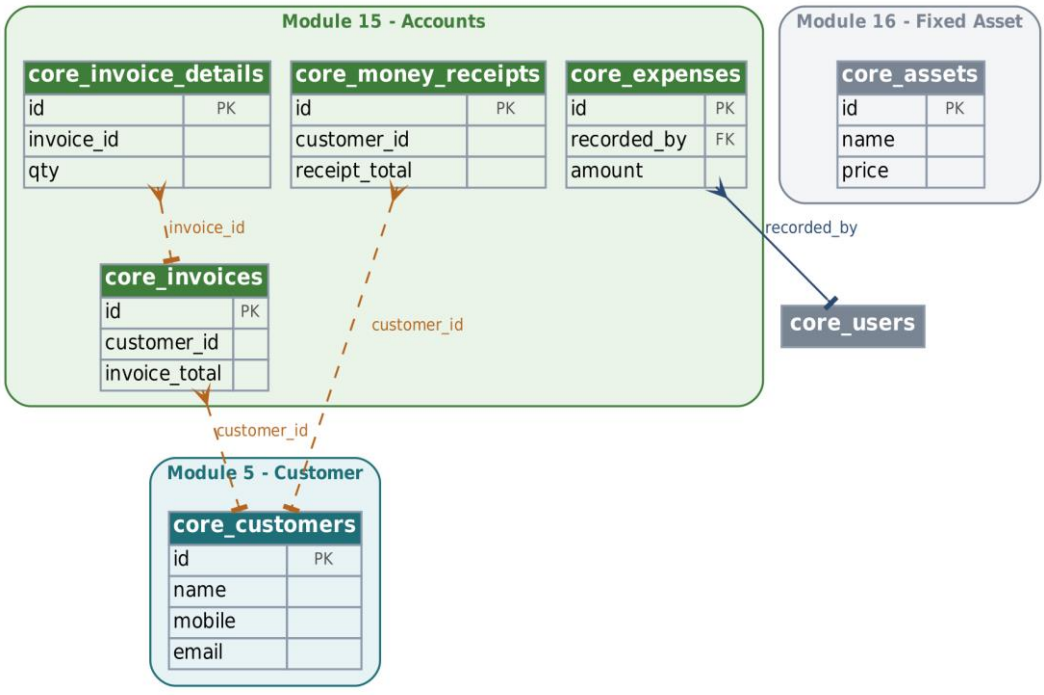


Figure 9: Invoices and money receipts belong to a customer; expenses are recorded by a user; the asset register is a standalone table.

6.9 Part 8 – Module 17: Shipment

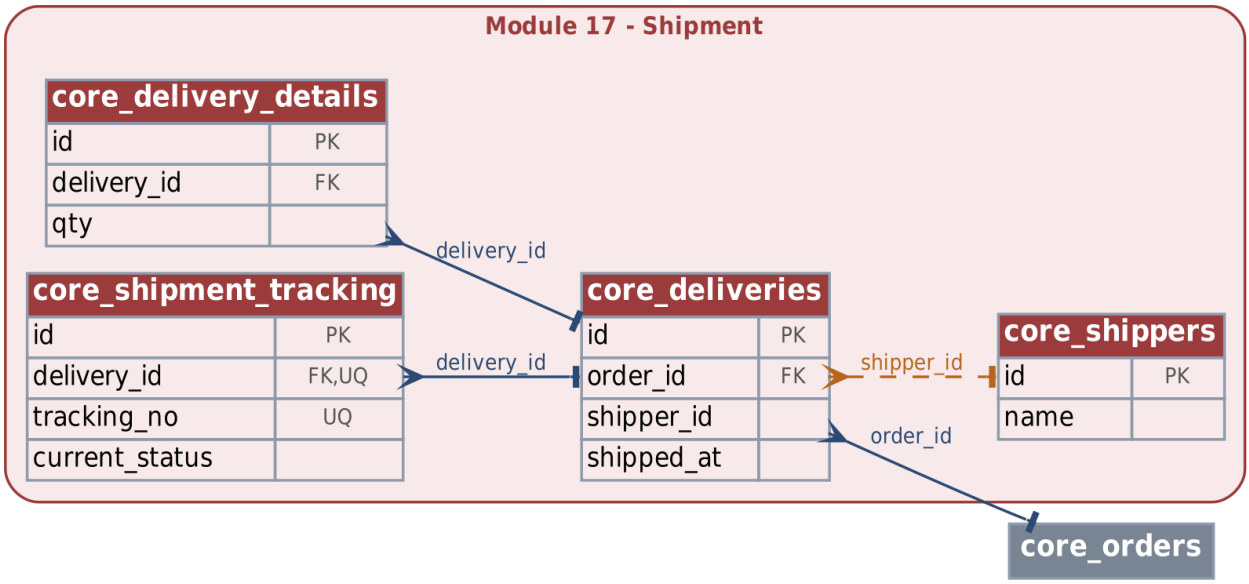


Figure 10: A delivery belongs to an order, has delivery details and a shipper, and is followed by one tracking record.

7. 30-Day Implementation Plan (Module-wise)

Day(s)	Module	Activities	Deliverable
1–2	#2 User & Role Management (+ project setup)	Import database, Core PHP MVC folders, login/logout, password hashing, roles, protected routes.	Login and role-based access
3	#3 Company & Department	Company profile, departments, positions, location data.	Master data screens
4	#4 Buyer, #5 Customer	Buyer and customer CRUD, validation, search.	Buyer and customer lists
5–6	#6 Merchandising	Style, costing, sample, buyer PO, size/color breakdown.	Buyer → style → PO demo
7–8	#7 Product & Order	Order header and line items, quantity, price, delivery date.	Order entry
9–10	#8 Catalogue & BOM	Products, categories, UOM, BOM header and raw materials.	Product and BOM screens
11–12	#9 Purchase	Suppliers, purchase order and items, supplier payment.	Purchase workflow
13–15	#10 Inventory	Warehouses, stock-in/out, adjustments, transaction history, balance.	Tested stock balance
16–18	#11 Production	Production order linked to order/BOM, stages, stage output.	Production progress
19–20	#12 Quality Control	Inspection, defect quantity, pass/fail linked to production.	QC workflow
21–22	#13 HR Management	Person and employee profile, department/position, daily attendance.	Employee and attendance
23	#14 Payroll (basic)	Payroll items and a simple payroll invoice per employee.	Basic payroll record
24–25	#15 Accounts	Invoice, money receipt, expense/income, outstanding calculation.	Finance summary
26	#16 Fixed Asset (basic)	Asset register CRUD.	Asset list
27	#17 Shipment	Delivery, shipper, tracking number, status.	Dispatch tracking
28	#1 Dashboard	Summary cards, recent orders, key counts.	Working dashboard
29	#18 Reports	Order, stock, production, QC, HR, shipment and accounts reports with filters.	Report set
30	Final integration	End-to-end test, FK/data validation, security check, README, SQL export, presentation.	Final demo and documents

Each module ends with a short test of its own workflow; Day 30 is reserved for integration testing and final documentation.

8. Assumptions and Exclusions

8.1 Assumptions

- Single company/factory and one base currency; sample data only, no live factory deployment.
- The instructor's approved requirements and evaluation rubric take precedence over this proposal.
- The current SQL is the database baseline; any schema change is versioned and documented.

8.2 Excluded from the 30-Day MVP

- Statutory payroll, tax, provident fund, overtime compliance; leave entitlement and shift automation.
- Advanced fabric inspection, lot traceability, cutting optimization, line balancing, machine/IoT integration.
- Barcode/RFID, mobile apps, external APIs, SMS/email and payment gateways.
- Full double-entry accounting, bank reconciliation, multi-company/multi-currency, complex approval workflows, asset depreciation.

9. Acceptance Criteria

- The application runs locally on XAMPP (PHP, MySQL/MariaDB); login/logout and server-side role checks work.
- Buyer, style, costing, buyer PO, order, product, BOM and purchase records can be created, viewed, edited and safely removed.
- Purchase, stock movement and adjustment records give consistent stock balances for test cases.
- Production orders, stage outputs and QC inspections are linked and validated; attendance can be maintained and summarized.
- Invoice totals, receipts, expenses and outstanding balances are calculated on the server; deliveries are tracked by status.
- Prepared statements, password hashing, input validation, output escaping and CSRF protection are applied.
- A database export, setup guide, user guide, test checklist, ERD and final presentation are delivered.

10. Risks and Mitigation

Risk	Mitigation
Scope exceeds 30 days	Deliver essential CRUD for every module first; keep payroll and assets basic; defer advanced features.
Duplicate or legacy tables	Choose canonical tables (core_users, core_attendance_daily) before building screens.
Missing foreign keys	Audit orphan records, then add constraints through versioned migrations.
Inconsistent quantity/money types	Standardize to DECIMAL after reviewing stored data.
Stock or financial calculation errors	Calculate on the server, use transactions, test with known sample cases.
Unauthorized access	Role checks, prepared statements, password hashing and session security.

11. Approval

This proposal defines the intended ERP blueprint and the practical 30-day trainee MVP. Final scope, priorities and any database changes remain subject to instructor review and approval.

Role	Name	Signature	Date
Trainee Name	JAKIR HOSEN LITON		
Instructor	Mohammad Towhidul Islam		
IsDB-BISEW Consultant/ Reviewer	Md. Moshaidul Islam		

— End of Comprehensive Garments ERP Proposal —